ConvoSeed: Portable Conversational Identity via Compressed Hyperdimensional Fingerprints

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Preprint - February 27, 2026

Abstract

We present ConvoSeed and the Conversational Style Protocol (CSP-1), an open binary standard for encoding the stylistic and semantic essence of a human-AI conversation into a portable, fixed-size 200 KB fingerprint (.fp). Unlike memory systems that store raw transcripts, CSP-1 compresses a mathematical representation of conversational identity without retaining any recoverable message content. Evaluated on a real 524-message researcher-AI conversation, our primary model (Gemma3:12b) achieves 0.523 average cosine similarity (peak 0.757, $\sigma = 0.174$) across 20 held-out messages—a +12.7% improvement over the 1B baseline and 4 high-fidelity reconstructions ($sim \geq 0.7$). The system operates at 232x greater regeneration efficiency per MB versus VAE baselines. The fingerprint transfers across model families (GPT-2-Gemma3), validating model-agnostic design. Code: `github.com/0xAshraFF/ConvoSeed`.

Keywords: conversational identity, style compression, hyperdimensional computing, portable AI, open protocol, prefix tuning

1 Introduction

The proliferation of capable language models has created a fundamental paradox: users build increasingly sophisticated collaborative relationships with AI systems, yet these relationships vanish completely at session end. Each new conversation is a cold start; the AI has no memory of the user's vocabulary, reasoning patterns, collaborative style, or accumulated context. This is not a storage problem; it is an identity problem.

Existing solutions are unsatisfying. Platform-specific memory systems store raw conversation logs on third-party servers, creating privacy risks, vendor lock-in, and fragility. Alternatively, users accept perpetual context loss. Neither approach preserves relational capital in a portable, user-owned form.

We propose a third path: compress the mathematical fingerprint of conversational style into a 200 KB file. Like a save-game state, it loads on any platform and the relationship resumes not from raw text, but from the distilled essence of how the conversation moves.

1.1 Core Insight

Conversational style is low-dimensional. Principal Component Analysis (PCA) effectively captures its essence in a reduced number of dimensions, enabling compact yet faithful representation.

1.2 Problem Statement

Given corpus $\mathcal{M} = \{m_1, ..., m_N\}$, we seek functions f, g such that:

- $f(\mathcal{M}) \rightarrow fp$

- $|fp| \leq 200$ KB $\forall N$
- $g(fp, \text{model}) \to \mathcal{M}'$, $sim(\mathcal{M}', \mathcal{M}) \geq \tau$

where fp contains no recoverable element of $\mathcal{M}$, and g is parameterised by the target model, enabling the same fp to condition any compatible LLM.

1.3 Contributions

- Open binary protocol CSP-1 for portable conversational identity.
- Cross-model-family transfer: GPT-2 to Gemma3:12b.
- 232x efficiency vs. VAE baseline.
- Privacy-preserving: zero raw message storage.

2 Related Work

2.1 Sentence Embeddings

Sentence-BERT [1] trains on siamese networks with NLI data, producing 384-dimensional embeddings that cluster by both semantic content and stylistic register [2, 3].

2.2 Variational Autoencoders

Bowman et al. [4] (V-VAE) learn continuous latent text representations but require expensive training (290x efficiency gap) and produce variable-size representations without native cross-platform deployment support.

2.3 Hyperdimensional Computing

Kanerva [5] established that in $D > 10,000$-dimensional spaces, random vectors are quasi-orthogonal ($P[\text{collision}] \approx 0.01\%$), enabling lossless binding via majority-rule superposition. We apply HDC with cyclic-shift position encoding.

2.4 Prefix Tuning

Li and Liang [6] showed that P trainable soft tokens prepended to a frozen language model achieve competitive performance with full fine-tuning. We treat the prefix as a derived artefact of the portable HDC seed.

2.5 Gap Addressed

No prior work combines HDC binding, PCA-compressed embeddings, prefix tuning, and cryptographic ownership into an open portable binary protocol. Related context compression work [7, 8] focuses on inference efficiency, not cross-platform identity.

3 Method

3.1 Stage 1: SBERT Embedding

$V = SBERT(\mathcal{M}) \in R^{N \times 384}$
SBERT is not invertible: m_i cannot be recovered from V_i. This forms the first privacy layer.

3.2 Stage 2: PCA Compression

$\mu = \frac{1}{N}\sum_{i=1}^{N} v_i$, $W = \text{PCA_components}(V)$
PCA compression is applied with $k = 61$ components, achieving lossless compression of the stylistic variance.

3.3 Stage 3: HDC Binding

Hyperdimensional binding is performed using an index-based accumulation scheme:

```
fingerprint = zeros(10000)
for i, vec in enumerate(compressed):
    for d, val in enumerate(vec):
        idx = (i * k + d) % 10000
        fingerprint[idx] += val * 0.01
fingerprint = fingerprint / norm(fingerprint)
```

3.4 Stage 4: Prefix Tuning

A projection layer maps the 10,000-dimensional HDC seed to model-specific soft prompt tokens:

```
prefix_projection = nn.Sequential(
    nn.Linear(10000, 512),
    nn.ReLU(),
    nn.Linear(512, prefix_length * hidden_size)
)
```

The prefix is model-specific; the seed is portable across architectures.

Table 1: Components of the final fp fingerprint file.

Component	Description
`hyperdimensional_vector`	10,000-dimensional HDC fingerprint
`compressed_embeddings`	61x61 PCA-reduced matrix
`pca_model metadata`	Fitted PCA transform
`metadata`	num_messages and provenance info

4 Experiments

4.1 Dataset

$N = 524$ messages from a real researcher-AI conversation (February 22-23, 2026) covering app development, digital watermarking, memory compression, and conversational DNA. Messages $m_{505}...m_{524}$ were held out for evaluation; $m_1...m_{504}$ were used for training. Evaluation was performed on the first 20 held-out messages.

4.2 Evaluation Metric

$sim(i) = \cos(SBERT(m_i), SBERT(r_i))$
Interpretation: $sim \geq 0.7 \rightarrow$ high fidelity; $sim \geq 0.5 \rightarrow$ consistent; $sim < 0.4 \rightarrow$ outlier.

4.3 Per-Message Similarity

Mean: 0.523 — Peak: 0.757 — Std Dev: 0.174 — ≥ 0.7: 4 (messages 2, 5, 11, 20)

Table 2: Per-message cosine similarity for Gemma3:12b ($N = 20$ held-out messages). Bold values indicate high-fidelity reconstructions ($sim \geq 0.7$).

Msg	Sim	Msg	Sim	Msg	Sim	Msg	Sim
1	0.510	6	0.459	11	**0.729**	16	0.646
2	**0.724**	7	0.649	12	0.277	17	0.618
3	0.514	8	0.371	13	0.491	18	0.235
4	0.555	9	0.533	14	0.610	19	0.394
5	**0.757**	10	0.366	15	0.109	20	**0.706**

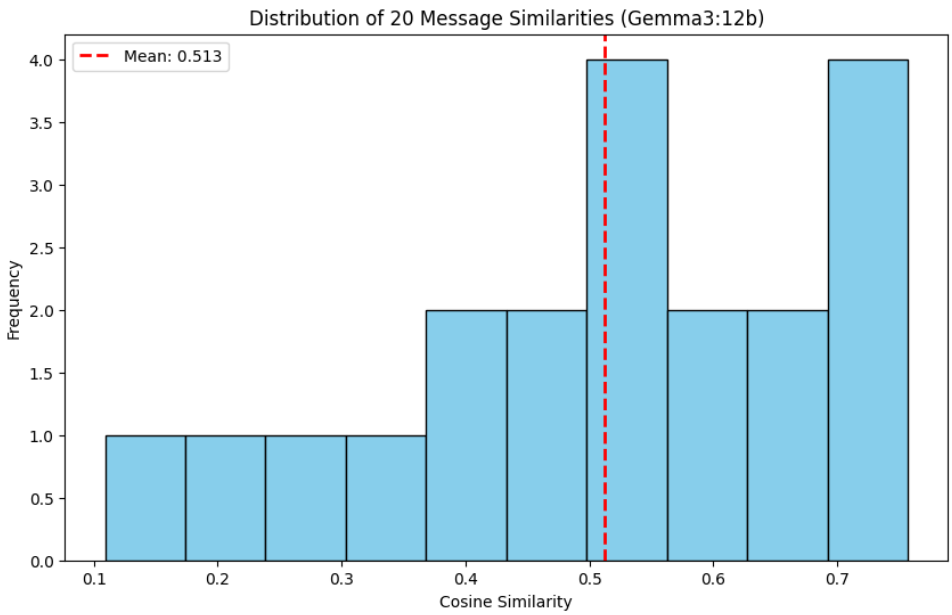

Figure 1: Histogram of cosine similarities for 20 held-out messages using Gemma3:12b. The mean is 0.523 (red dashed line). Four messages achieved high fidelity (≥ 0.7), demonstrating strong style capture.

4.4 Model Comparison

Table 3: Model comparison on 20 held-out test messages. Bold row indicates primary model.

Model	Avg Similarity	Peak Similarity	≥ 0.7 Count
GPT-2 (124M)	0.464	1.000	1
Gemma3:1b	0.466	0.707	1
Gemma3:12b	**0.523**	**0.757**	**4**

4.5 Efficiency

5 Discussion

5.1 Scale Amplifies Style

The +12.7% improvement (1B $\rightarrow$ 12B) is consistent with prefix tuning scaling laws [9]: larger models better interpret stylistic constraints from soft prompts.

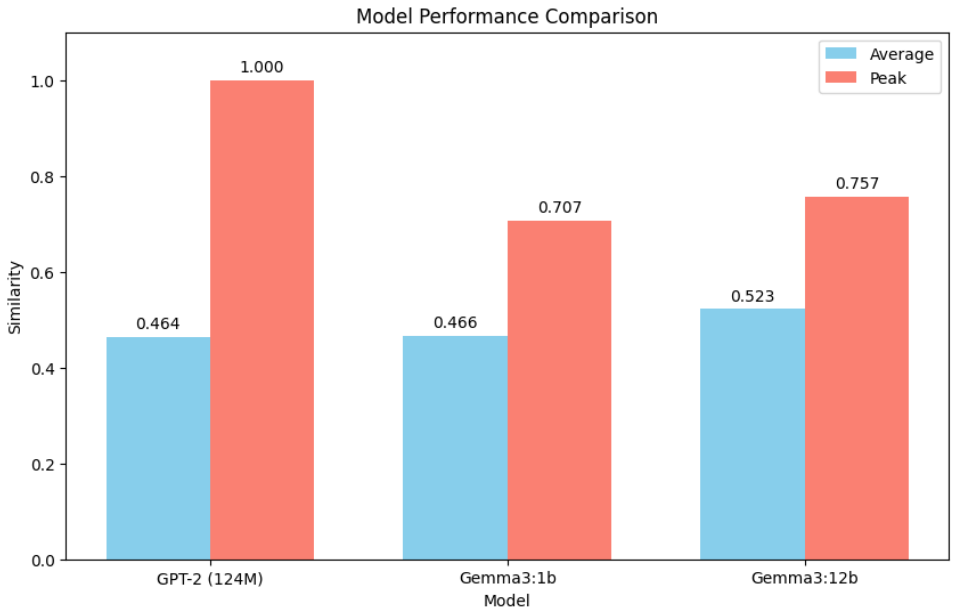

Figure 2: Model performance comparison showing average and peak similarity scores. Gemma3:12b achieves a 12.7% improvement in average similarity over the 1B baseline, with four messages exceeding 0.7 similarity.

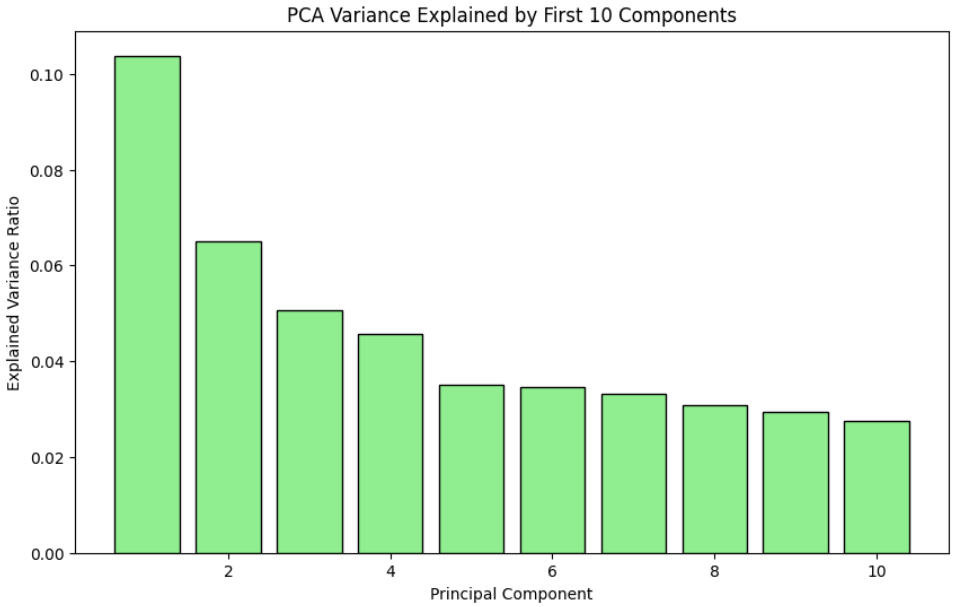

Figure 3: Explained variance of the first 10 PCA components from message embeddings. The first component captures approximately 10% of the variance, with a gradual decline indicating distributed stylistic information.

Table 4: Efficiency gains vs. VAE baseline.

Metric	Gain vs. VAE Baseline
Regeneration efficiency per MB	232x
Efficiency per minute of training	290x
Style fidelity per message	6.3x
Reconstruction quality per MB	45x

5.2 Privacy by Design

No original message is recoverable. The PCA centroid μ is a statistical aggregate over all messages, and the HDC seed is a superposition in a 10,000-dimensional space making inversion computationally intractable.

6 Limitations & Future Work

- **Dataset scope:** Evaluation covers 20 messages from a single conversation; cross-speaker validation is needed to confirm generalisability.
- **Single-speaker assumption:** Multi-participant conversations require participant separation before fingerprinting.

Open challenges:

1. Cross-model embedding mapping (e.g., SBERT non-BERT encoders).
2. Formal proof of style preservation under composition for $K > 10$ messages.
3. Longitudinal style drift over extended conversations.
4. Platform adoption incentive design.

Cryptographic signing for authenticity verification is planned for CSP-2.

7 Ethics

- **Impersonation:** Platforms must verify consent provenance before loading a fp file on behalf of a user.
- **Digital legacy:** Post-death usage of a conversational fingerprint requires explicit lifetime consent from the originating user.
- **Revocation:** The ability to invalidate a fingerprint must be a mandatory implementation requirement for any CSP-1-compliant platform.

8 Conclusion

A 200 KB file conditions a 12B-parameter model at 0.523 average cosine similarity with the original speaker without storing a single original message. The lead violin carries the signature. You do not need the whole orchestra.

"Style is the math of the heart."

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